Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **automated test frameworks**, **distributed systems**, and **performance-optimized pipelines** across multi-platform environments. Experienced in C++, Python, and systems-level programming with strong testing and CI/CD practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Computer Networks

Experience

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Designed **automated test framework** with 20+ test suites using PyTest and Mockito, implementing nightly automation pipelines for build validation, integration testing, and regression testing across cross-platform environments.
- Built **VM-based test automation** for executable deployment across Windows, Linux, and macOS, automating build compilation, installation, and end-to-end validation with Jenkins CI/CD and CSV report generation.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **high-performance data pipelines** processing 100K+ sensor trajectory records, implementing efficient feature extraction and batch processing workflows optimized for memory and CPU constraints.
- Designed **reproducible experimentation framework** for model training and evaluation, achieving 18% improvement in planning accuracy through systematic benchmarking and performance optimization.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Optimized **system performance** for 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexing strategies, and efficient data access patterns across distributed service architecture.
- Built **automated validation pipelines** processing 5K+ monthly jobs with error handling, reconciliation checks, and monitoring, achieving 99%+ reliability through systematic testing and deployment workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **distributed real-time synchronization system** handling 2M+ records across locations, implementing WebSocket-based communication with conflict resolution and low-latency data propagation.
- Automated **build and deployment pipelines** with Docker and Jenkins, cutting release cycles by **50%** through containerized testing, staged deployments, and systematic validation workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **concurrent batch processing system** using ThreadPoolExecutor, optimizing throughput for high-volume workloads while maintaining stable performance under resource constraints.
- Developed **modular backend architecture** with Blueprint-based design, enabling maintainable code structure, systematic testing, and efficient debugging across production deployments.

Skills

Languages: **C++**, **Python**, Java, SQL, JavaScript/TypeScript, Bash
Systems & Testing: Test automation, PyTest, Mockito, Jest, unit/integration testing, CI/CD pipelines
Distributed Systems: Microservices, real-time sync, concurrent processing, performance optimization
Tools & Platforms: Git, Linux, Docker, Jenkins, AWS, Azure, VM automation
Fundamentals: Data structures, algorithms, OS internals, resource scheduling, memory management

Projects

CPU Scheduling Simulator Live | *C++, React, JavaScript*

- Built a **systems-level scheduling simulator** in C++ implementing FCFS, SJF, SRT, Round Robin, and MLQ algorithms, optimizing resource allocation and reducing average turnaround time by **25%**.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized processing pipeline** on AWS ECS/Fargate with automated validation gates, implementing systematic testing and deployment workflows for production-grade reliability.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **scalable distributed platform** supporting 50K+ users and 1M+ transactions with Kubernetes orchestration, real-time synchronization, and systematic performance monitoring.